Task - 2

Arrays Manipulation.

Aim:- To write a Java Program that Calculate the prefix sum of an array where each elements in the prefix index of i.

Algorithm :-

1. Start the program
2. Read the size of the array N.
3. Read N elements into the array
4. Create another array prefixSum of the same size.
5. Set prefixSum[0] = arr[0].
6. Traverse the array from index 1 to n-1
   - Compute prefixSum[i] = prefixSum[i-1] + arr[i]
7. Display the prefix sum array.
8. Stop the program.

Program

```
import java.util.Scanner;
Class PrefixSumArray {
    public static void main(String[] args) {
```

output

Enter the size of the array = 5

Enter the array elements

10 20 10 5 15

prefix sum array:

10 30 40 45 60

```
Scanner sc= new Scanner (System.in);
System.out.println ("Enter the size of
                    the array :");
  int n = sc.nextInt();
  int[] arr = new int [n];
  int [] prefixsum  =new int [n];
  System.out.println ("Enter the array
                    elements :");
  for (int i =0; i<n; i++){
      arr[i] = sc.nextInt();
  }
// Calculate prefix sum
  prefixsum[0] = arr[0];
  for (int i =1; i<n; i++) {
      Prefix sum[i] =prefix sum[i-1]+ arr[i];
  }
  //Display prefix sum array.
  System.out.println ("prefixsum ~~[i-1]~~ +arr
                    sum arr[i]");
  for (int sum: prefix sum){
      System.out.println (num +" ");
  }
}
}
```

	2
	5
5	5
5	5
5	5
20	20
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Result:- Thus, the Java program successfully compute the prefix sum array by adding elements from the begining of the array up to each index